

Methun Raj

AI & Automation Engineer

Hamburg, Germany · +49 1577 3686620 · methunrajmani85@gmail.com

[linkedin.com/in/methun-raj-0335b617b](https://www.linkedin.com/in/methun-raj-0335b617b)

Permanent residence · Unrestricted work authorisation



PROFILE

Data engineer building production LLM systems rather than prototypes. Designed and operate an AI document extraction platform in daily production use, with multi-provider model routing and per-project cost control. Previously forecast the semiconductor, cloud and AI markets at Statista. Current focus: agentic workflows, and how you evaluate what an agent produced before it ships.

PROFESSIONAL EXPERIENCE

Data Engineer

Jan 2026 – Present

Statista GmbH, Hamburg · Full-time

- Build ingestion and retrieval pipelines for real-time news intelligence: GDELТ ingest, deduplication, embedding generation.
- Develop semantic search over large sharded vector indexes, plus a Python client library for agentic retrieval and bulk export.
- Own pipeline reliability — observability and alerting, so failures surface instead of passing silently.

AI Consultant & AI Automation Engineer

Mar 2025 – Present

Swift ProSys · Freelance, Remote

- Built and operate an AI document extraction platform used by **~200 people** across language-specific teams, processing **6,000,000,000 tokens** in five months.
- Route workloads across **five model providers** — OpenAI, Google Gemini, Qwen, Vector Engine and local Ollama — behind one interface, each with correct billing for reasoning, cached and thinking tokens.
- Designed the reliability layer: a 12-category failure taxonomy, a four-attempt retry ladder that escalates prompt strictness and token budget, and non-blocking backoff that never occupies a concurrency slot.
- Encrypted provider credentials server-side with per-project isolation; per-project cost analytics make model spend predictable before a run.
- Maintain a 100-test suite (Vitest and Deno) with linting and dead-code analysis in CI.

Data Analyst

Apr 2025 – Jan 2026

Statista GmbH, Hamburg

- Produced forecasts and KPI models for the Semiconductors, Public Cloud, AI and Data Center segments of Statista's **Technology Outlook**.
- Automated recurring analysis, improving turnaround across release cycles.

Junior Analyst

Dec 2023 – Mar 2025

Statista GmbH, Hamburg

- Researched and published market size, revenue and growth KPIs across technology segments.

Working Student

Jul 2022 – Oct 2023

Statista GmbH, Hamburg

- First Python and SQL pipelines; product matching with applied NLP.

Technical Intern

Jul 2020 – Jan 2021

Garuda Aerospace, Chennai, India

- Statistical analysis of drone telemetry; identified the drivers of battery performance.

EDUCATION

M.Sc. Smart Systems

Oct 2021 – Oct 2023

Hochschule Furtwangen University, Germany

- Thesis: a convolutional network identifying regional plant and insect species from photographs, served behind an API with continuous retraining.

B.E. Mechatronics Engineering

2017 – 2021

Agni College of Technology, Chennai, India · Grade 8.27 / 10

SKILLS

Programming	Python, SQL, TypeScript
AI & LLM	Large Language Models (LLM), Retrieval-Augmented Generation (RAG), vector search and vector databases, embeddings, AI agents and agentic workflows, LangChain, LangGraph, Agno, prompt engineering, model evaluation, OpenAI API, Google Gemini API, vLLM, Ollama, MCP
Data & Cloud	AWS (S3, Lambda), Docker, PostgreSQL, pgvector, Supabase, ETL and data pipelines, data modelling, observability, CI/CD
Analysis	Market sizing and forecasting, KPI modelling, competitive benchmarking, pandas, NumPy
Tooling	Claude Code, OpenAI Codex, Cursor, GitHub Actions, Sentry

LANGUAGES

English — full professional proficiency · German — B1, working towards B2

CERTIFICATIONS

Python for Data Analysis, Coursera (2022) · Machine Learning with Python, Coursera (2022)